

Домашняя работа по дифференциальным уравнениям

• (N832)
$$\begin{cases} \dot{x} = 2y - x + 1 \\ \dot{y} = 3y - 2x \end{cases}$$

$$A = \begin{pmatrix} -1 & 2 \\ -2 & 3 \end{pmatrix}$$

$$\det(A - \lambda E) = \begin{vmatrix} -1-\lambda & 2 \\ -2 & 3-\lambda \end{vmatrix} = -(1+\lambda)(3-\lambda) + 4 =$$

$$= -3 - 2\lambda + \lambda^2 + 4 = (\lambda - 1)^2$$

$$\lambda_{1,2} = 1 \quad A - \lambda_1 I = \begin{pmatrix} -2 & 2 \\ -2 & 2 \end{pmatrix} \sim \begin{pmatrix} -2 & 2 \\ 0 & 0 \end{pmatrix} \quad e_1 = (1, 1)^T$$

$$L = h$$

$$x = (h_1 + j_1 t) e^t$$

$$y = (h_2 + j_2 t) e^t$$

$$\begin{cases} j_1 + h_1 + j_1 t = 2h_2 + 2j_2 t - h_1 - j_1 t \\ j_2 + h_2 + j_2 t = 3h_2 + 3j_2 t - 2h_1 - 2j_1 t \end{cases}$$

$$\begin{cases} j_1 + h_1 + j_1 t = 2h_2 + 2j_2 t - h_1 - j_1 t \\ j_2 + h_2 + j_2 t = 3h_2 + 3j_2 t - 2h_1 - 2j_1 t \end{cases}$$

$$x = (c_1 + 2c_2 t) e^t$$

$$x_{21} = A$$

$$0 = 2B - A + 1$$

$$y = (c_1 + c_2 + 2c_2 t) e^t$$

$$y_{21} = B$$

$$0 = 3B - 2A$$

$$A = 2B + 4$$

$$0 = 3B - 4B - 2$$

$$B = -2, A = -3$$

$$x = (c_1 + 2c_2 t)e^t - 3$$

$$y = (c_1 + c_2 + 2c_2 t)e^t - 2$$

$$\text{Omkem: } \begin{cases} x = (c_1 + 2c_2 t)e^t - 3 \\ y = (c_1 + c_2 + 2c_2 t)e^t - 2 \end{cases}$$

N 633 Решите линейную неогомогенную систему

$$\begin{cases} \dot{x} = 2x + y + e^t \\ \dot{y} = -2x + 2t \end{cases}$$

$$A = \begin{pmatrix} 2 & 1 \\ -2 & 0 \end{pmatrix}$$

$$\det(A - \lambda E) = \begin{vmatrix} 2-\lambda & 1 \\ -2 & -\lambda \end{vmatrix} =$$

$$= -\lambda(2-\lambda) + 2 = 0$$

$$-\lambda + \lambda^2 + 2 = 0$$

$$D = 4 - 4 \cdot 2 = -4$$

$$\lambda_{1,2} = \frac{2 \pm 2i}{2} = 1 \pm i$$

$$A - \lambda_1 I = \begin{pmatrix} 2-1-i & 1 \\ -2 & -1-i \end{pmatrix} = \begin{pmatrix} 1-i & 1 \\ -2 & -1-i \end{pmatrix} \xrightarrow{1+i} \sim$$

$$\sim \begin{pmatrix} 1+i & 1+i \\ -2 & -1-i \end{pmatrix} \sim \begin{pmatrix} 2 & 1+i \\ 0 & 0 \end{pmatrix}$$

$$e_1 = (-2, 1+i)^T$$

$$e_2 = (2, 1+i)^T$$

$$X_1 = 2e^{(1+i)t} = 2e^t(\cos t + i \sin t)$$

$$X = 2e^t(c_1 \cos t + c_2 \sin t)$$

$$\begin{aligned} y_1 &= (1+i)e^{(1+i)t} = (1+i)e^t(\cos t + i \sin t) \\ &= e^t(\cos t + i \sin t + i \cos t - \sin t) = \\ &= e^t(\cos t - \sin t + i \sin t + i \cos t) \end{aligned}$$

$$y = e^t[c_1(\cos t - \sin t) + c_2(\sin t + \cos t)]$$

$$\begin{cases} \dot{x} = A e^t \\ \dot{y} = B e^t \end{cases}$$

$$\begin{cases} A = 2A + B + 2 \\ B = -2A \end{cases}$$

$$\Rightarrow \begin{cases} A = 1 \\ B = -2 \end{cases}$$

Решение: $\begin{cases} \ddot{x} = e^t \\ \ddot{y} = -2e^t \end{cases}$

$$\begin{cases} x_{12} = At + B \\ y_{12} = Ct + D \end{cases} \quad \begin{cases} a = 2at + 2b + et + d \\ c = -2at - 2b + 2t \end{cases}$$

Решая систему, найдем:

$$\begin{cases} a = 2b + d \\ 0 = 2a + c \\ c = -2b \\ 0 = -2a + 2 \end{cases} \quad \begin{matrix} a = 1 \\ c = -2 \\ b = 1 \\ d = -1 \end{matrix} \quad \rightarrow \begin{cases} x_{12} = t + 1 \\ y_{12} = -2t - 1 \end{cases}$$

откуда:

$$\begin{cases} x = 2e^t c_1 \cos t + 2e^t c_2 \sin t + et + t + 1 \\ y = c_1 e^t (\cos t - \sin t) + c_2 e^t (\sin t + \cos t) - 2e^t - 2t - 1 \end{cases}$$

№836. Решить линейную неоднородную систему

$$\begin{cases} \dot{x} = 2x - y \\ \dot{y} = y - 2x + 10t \end{cases} \quad A = \begin{pmatrix} 2 & -1 \\ -2 & 1 \end{pmatrix}$$

$$\det(A - \lambda E) = \begin{vmatrix} 2-\lambda & -1 \\ -2 & 1-\lambda \end{vmatrix} = (2-\lambda)(1-\lambda) + 2 = 0$$

$$\lambda_1 = 0, \quad \lambda_2 = 3$$

для $\lambda=0$

$$A - \lambda_1 I = \begin{pmatrix} 2 & -1 \\ 0 & 0 \end{pmatrix}$$

$$2L_1 = \beta$$

$$L_1 = \frac{\beta}{2}$$

$$e_1 = \left(\frac{\beta}{2}, \beta \right)^T$$

при $\beta=2$ $e_1 = (1, 2)^T$

для $\lambda=3$

$$A - \lambda_2 I = \begin{pmatrix} -1 & -1 \\ -2 & -2 \end{pmatrix} \sim \begin{pmatrix} -1 & -1 \\ 0 & 0 \end{pmatrix}$$

$$L_2 = -\beta$$

$$e_2 = (-\beta, \beta)^T$$

при $\beta=-1$ $e_2 = (1, -1)^T$

$$\begin{cases} x = at^2 + bt + c \\ y = dt^2 + et + f \end{cases} \quad \begin{cases} 2at + b = 2at^2 + 2bt + 2c - dt^2 - et - f \\ 2dt + e = -2at^2 - 2bt - 2c + dt^2 + et + f \end{cases}$$

Составим и решим систему (конкретно состав. коэффициентов)

$$2a - d = 0$$

$$e = -2c + f$$

$$2a = 2b - e$$

$$b = 2c - f$$

$$-2a + d = 0$$

$$2d = -2b + e + f$$

$$\begin{cases} 2a = d \\ e = 2b - 2a \Rightarrow 2a = 2b - e \\ f = 2c - b \\ e = 2d + 2b - 10 \\ f = e + 2c \\ e = f - 2c \end{cases} \Rightarrow \begin{cases} d = 6 \\ a = 3 \\ b = 2 \\ e = -2 \\ c = 0 \\ f = -2 \end{cases}$$

Particular: $\begin{cases} x = C_1 e^{3t} + C_2 + 3t^2 + 2t \\ y = -C_1 e^{3t} + 2C_2 + 6t^2 - 2t - 2 \end{cases}$

Orbital: $\begin{cases} x = C_1 e^{3t} + C_2 + 3t^2 + 2t \\ y = -C_1 e^{3t} + 2C_2 + 6t^2 - 2t - 2 \end{cases}$

(N839) $\begin{cases} \dot{x} = 2x - 3y \\ \dot{y} = x - 2y + 2\sin t \end{cases}$

$$A = \begin{pmatrix} 2 & -3 \\ 1 & -2 \end{pmatrix} \quad F = \begin{pmatrix} 0 \\ 2\sin t \end{pmatrix}$$

$$\det(A - \lambda E) = \begin{vmatrix} 2-\lambda & -3 \\ 1 & -2-\lambda \end{vmatrix} = -(2+\lambda)(2-\lambda) + 3 = -(4-\lambda^2) + 3 = \lambda^2 - 1 = 0 \Rightarrow \lambda_{1,2} = \pm 1$$

$$\text{find } \lambda_1 = -1$$

$$A - \lambda_1 I = \begin{vmatrix} 3 & -3 \\ 1 & -1 \end{vmatrix} \sim \begin{vmatrix} 0 & 0 \\ 1 & -1 \end{vmatrix} \quad e_1 = (1, 1)^T$$

$$\text{find } \lambda_2 = 1$$

$$A - \lambda_2 I = \begin{vmatrix} 1 & -3 \\ 1 & -3 \end{vmatrix} \sim \begin{vmatrix} 1 & -3 \\ 0 & 0 \end{vmatrix} \quad e_2 = (3, 1)^T$$

$$P = \begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix} \quad J(t) = e^{Jt}$$

$$\bar{y}(t) = \begin{pmatrix} 1 & 3 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} e^{-t} & 0 \\ 0 & e^t \end{pmatrix} = \begin{pmatrix} e^{-t} & 3e^t \\ e^{-t} & e^t \end{pmatrix}$$

$$\bar{y}^{-1}(t) F(t) = -\frac{1}{2} \begin{pmatrix} e^t & -3e^t \\ -e^{-t} & e^t \end{pmatrix} \begin{pmatrix} 0 \\ 2\sin t \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} -6e^t \sin t \\ 2e^{-t} \sin t \end{pmatrix}$$

$$= \begin{pmatrix} 3e^t \sin t \\ -e^{-t} \sin t \end{pmatrix}$$

$$C_1 = \int 3e^t \sin t \, dt = \frac{3e^t \sin t - 3e^t \cos t}{2} + \tilde{C}_1$$

$$C_2 = \int -e^{-t} \sin t \, dt = \frac{\cos t + \sin t}{2e^t} + \tilde{C}_2$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = C_1 \begin{pmatrix} 3 \\ 1 \end{pmatrix} e^t + C_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^{-t}$$

$$x = a \sin t + b \cos t$$

$$y = c \sin t + d \cos t$$

$$\dot{x} = a \cos t - b \sin t$$

$$\dot{y} = c \cos t - d \sin t$$

$$\begin{cases} a \cos t - b \sin t = (2a - 3c) \sin t + (2b - 3d) \cos t \end{cases}$$

$$\begin{cases} c \cos t - d \sin t = (a - 2c + 2) \sin t + (b - 2d) \cos t \end{cases}$$

Permanen equality:

$$\begin{cases} a = 2b - 3d = -d + 2c - 2 \end{cases}$$

$$\begin{cases} -b = (2a - 3c) \end{cases}$$

$$\begin{cases} c = b - 2d \Rightarrow b = c + 2d \end{cases}$$

$$-d = a - 2c + 2 \Rightarrow a = -d + 2c - 2$$

$$2b = 2d + 2c - 2$$

$$b = d + c - 1$$

$$c + 2d = d + c - 1$$

$$d = -1$$

$$a = -1 + 2c$$

$$b = c - 2$$

$$c = 3c - 2a$$

$$b = 6$$

$$c - 2 = 3c - 2a \Rightarrow a = 3$$

$$c - 2 = 3c + 2 - 4c \Rightarrow 2c = 4 \Rightarrow c = 2$$

$$\begin{cases} d = -1 \\ b = 6 \\ a = 3 \\ c = 2 \end{cases}$$

Answer: $\begin{cases} x = 3 \sin t \\ y = 2 \sin t - \cos t \end{cases}$

$\frac{2+2i}{2+2i} = 0$

General solution:

$$\begin{cases} x = 3c_1 e^t + c_2 e^{-t} + 3\sin t \\ y = c_1 e^t + c_2 e^{-t} + 2\sin t - \cos t \end{cases}$$

Orbital: $\begin{cases} x = 3c_1 e^t + c_2 e^{-t} + 3\sin t \\ y = c_1 e^t + c_2 e^{-t} + 2\sin t - \cos t \end{cases}$

(N844)

$$\begin{cases} \dot{x} = x - y + 1 \\ \dot{y} = 5x - y \end{cases}$$

$$A = \begin{pmatrix} 1 & -1 \\ 5 & -1 \end{pmatrix}$$

$$F(t) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{vmatrix} 1-\lambda & -1 \\ 5 & -1-\lambda \end{vmatrix} = (1-\lambda)(-1-\lambda) + 5 = 0 \Rightarrow$$

$$\Rightarrow \lambda_{1,2} = \pm 2i$$

$$\begin{aligned} A - \lambda E &= \begin{pmatrix} 1-2i & -1 \\ 5 & -1-2i \end{pmatrix} \sim \begin{pmatrix} 1-2i & -1 \\ 1 & -1-2i \end{pmatrix} \sim \\ &\sim \begin{pmatrix} 0 & \frac{-(1-2i)(1+2i)}{5} - 1 \\ 1 & -1-2i \end{pmatrix} \sim \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \end{aligned}$$

$$(1-2i)x_1 = x_2$$

$$x_1 = 1$$

$$x_2 = 1-2i$$

$$p_1 = (1, 1-2i)^T$$

$$p_2 = \bar{p}_1 = (1, 1+2i)^T$$

$$P = \begin{pmatrix} 1 & 1 \\ 1-2i & 1+2i \end{pmatrix} e^{JA t} \begin{pmatrix} e^{2it} & 0 \\ 0 & e^{-2it} \end{pmatrix}$$

$$g(t) = P e^{JA t} C(t) = \begin{pmatrix} 1 & 1 \\ 1-2i & 1+2i \end{pmatrix} \begin{pmatrix} e^{2it} & 0 \\ 0 & e^{-2it} \end{pmatrix} C(t) =$$

$$= \begin{pmatrix} e^{2it} & e^{-2it} \\ (1-2i)e^{2it} & (1+2i)e^{-2it} \end{pmatrix} C(t)$$

$$\tilde{y}^{-1}(t) = \frac{1}{4i} \begin{pmatrix} (1+2i)e^{-2it} & -e^{-2it} \\ (2i-1)e^{2it} & e^{2it} \end{pmatrix}$$

$$\tilde{y}^{-1} F = \frac{1}{4i} \begin{pmatrix} (1+2i)e^{-2it} & -e^{-2it} \\ (2i-1)e^{2it} & e^{2it} \end{pmatrix} \begin{pmatrix} 2t \\ 0 \end{pmatrix} = \frac{1}{i} \begin{pmatrix} (2t+1+2i)e^{-2it} \\ 2t(2i-1)e^{2it} \end{pmatrix}$$

$$C_1 = -2i(1+2i) \int t e^{-2it} dt = (1+2i) \left[t e^{-2it} + \frac{1}{2i} e^{-2it} \right] + C_1$$

$$C_2 = -2i(2i-1) \int t e^{2it} dt = (1-2i) \left[t e^{2it} - \frac{1}{2i} e^{2it} \right] + C_2$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = C_1 \begin{pmatrix} 1 \\ 1-2i \end{pmatrix} e^{2it} + C_2 \begin{pmatrix} 1 \\ 1+2i \end{pmatrix} e^{-2it}$$

$$\bar{y}(t) = \begin{pmatrix} e^{2it} & e^{-2it} \\ (1-2i)e^{2it} & (1+2i)e^{2it} \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = C_1 \begin{pmatrix} 1 \\ 1-2i \end{pmatrix} e^{2it} + C_2 \begin{pmatrix} 1 \\ 1+2i \end{pmatrix} e^{-2it}$$

$$\begin{cases} x = at^2 + bt + c \\ y = dt^2 + et + f \end{cases}$$

$$\begin{cases} \dot{x} = 2at + b = 2at^2 + bt + c - dt^2 - et - f + 8t \\ \dot{y} = 2dt + e = 5at^2 + 5bt + 5c - dt^2 - et - f \end{cases}$$

$$\begin{cases} a - d = 0 \\ 2a = b - e + 8 \\ b = c - f \\ 2d = 5b - e \\ e = 5c - f \end{cases} \Rightarrow \begin{cases} a = d \\ e = b - 2a + 8 \\ f = c - b = c - 2 \\ e = 5b - 2d = 5b - 2a \\ e = 5c - f = 10 - 2a \end{cases}$$

$$5a = d \quad 5b - 2a = b - 2a + 8$$

$$4b = b \\ b = 2$$

$$\begin{aligned} a = d = 0 &= f \\ e &= 10 \\ c &= 2 \end{aligned}$$

$$\begin{cases} x = C_1 e^{2it} + C_2 e^{-2it} + 2t + 2 \\ y = (1-2i)C_1 e^{2it} + (1+2i)C_2 e^{-2it} + 10t \end{cases}$$

Ответ:
$$\begin{cases} x = C_1 e^{2it} + C_2 e^{-2it} + 2t + 2 \\ y = (1-2i)C_1 e^{2it} + (1+2i)C_2 e^{-2it} + 10t \end{cases}$$

№45 Задача неоднородного линейного уравнения

$$\begin{cases} x' = 2x - y \\ y' = 2y - x - 5e^t \sin t \end{cases}$$

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \quad f = \begin{pmatrix} 0 \\ -5e^t \sin t \end{pmatrix}$$

$$\det(A - \lambda E) = \begin{vmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{vmatrix} = (2-\lambda)^2 - 1 = 0$$

$$\lambda^2 - 4\lambda + 3 = 0$$

$$D = 16 - 4 \cdot 3 = 4$$

$$\lambda_{1,2} = \frac{4 \pm 2}{2} = 3, 1$$

при $\lambda_1 = 3$

$$A - \lambda_1 I = \begin{pmatrix} -1 & -1 \\ -1 & -1 \end{pmatrix} \sim \begin{pmatrix} -1 & -1 \\ 0 & 0 \end{pmatrix} \quad e_1 = (1, 1)^T$$

$$-1 = 0 \quad 1 = -1$$

при $\lambda_2 = 1$ $A - \lambda_2 I = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \sim \begin{pmatrix} 1 & -1 \\ 0 & 0 \end{pmatrix} \quad e_2 = (1, 1)^T$

$$P = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \quad J(t) = P e^{Jt}$$

$$g(t) = \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} e^{3t} & 0 \\ 0 & e^t \end{pmatrix} = \begin{pmatrix} e^{3t} & e^t \\ -e^{3t} & e^t \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = c_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix} e^t$$

$$\begin{cases} x = a e^t \sin t + b e^t \cos t \\ y = c e^t \sin t + d e^t \cos t \end{cases}$$

$$\begin{cases} \dot{x} = a e^t \cos t - b e^t \sin t = 2a e^t \sin t + 2b e^t \cos t - c e^t \\ \dot{y} = c e^t \cos t - d e^t \sin t = -a e^t \sin t - b e^t \cos t + c e^t \sin t + d e^t \cos t + 2c e^t \sin t + 2d e^t \cos t - 5e^t \sin t \end{cases}$$

$$\begin{cases} a + b = 2b - 1 \\ a - b = 2a - c \\ b + c = -b + 2d \\ c - d = -a + 2c - 5 \end{cases}$$

$$\Rightarrow \begin{cases} a = \frac{3}{2} - 1 \\ b = 2 \\ c = 1 \\ d = \frac{3}{2} \end{cases}$$

~~Ansatz:~~

$$\begin{cases} x = c_1 e^{3t} + c_2 e^t + \frac{1}{2} e^t \sin t + e^t \\ y = -c_1 e^{3t} + c_2 e^t + 2e^t \sin t + \frac{3}{2} e^t \cos t \end{cases}$$

Ansatz:

$$\begin{cases} x = c_1 e^{3t} + c_2 e^t - e^t \sin t + 2e^t \cos t \\ y = -c_1 e^{3t} + c_2 e^t + e^t \sin t + 3e^t \cos t \end{cases}$$

Ansatz:

$$\begin{cases} x = c_1 e^{3t} + c_2 e^t - e^t \sin t + 2e^t \cos t \\ y = -c_1 e^{3t} + c_2 e^t + e^t \sin t + 3e^t \cos t \end{cases}$$

$\sin t \rightarrow e^t \cos t$